Online Medication & Prescription Tracker

A smart system to manage medicines, prescriptions, and pharmacy stock

Internship Project Presentation

Problem & Project Overview

Why this project was built

The Problem

- Patients forget to take medicines on time
- Hard to track multiple prescriptions
- Pharmacies run out of stock without warning
- Doctors cannot check if patients follow prescriptions
- No single platform for all — patient, doctor, pharmacy

Our Solution

- One app for Patients, Doctors, Pharmacists & Admin
- Doctors issue digital prescriptions easily
- Patients get reminders so they never miss a dose
- Pharmacies get alerts when stock is low
- Dashboard shows reports and health trends

Milestone 1 — Weeks 1 & 2

Authentication & Role Setup

Week 1 — Setup

- Set up the project folder
- Configure Spring Boot & React
- Design the database tables

Week 2 — Login & Roles

- Build login & signup pages
- Add JWT token for security
- Encrypt user passwords

Role Permissions

- Patient — view prescriptions
- Doctor — issue prescriptions
- Pharmacist — manage stock
- Admin — see everything

User Profile Setup

- Patients fill in their medical history and contact details
- Doctors provide their medical license number and specialization
- Pharmacists add their shop name and location details
- All profile info is saved securely and used to personalize the experience

Milestone 2 — Weeks 3 & 4

Prescription Management

Week 3 — Create Prescriptions

- Doctors fill a form with medicine name, dosage, duration
- Or they can upload a PDF prescription directly
- Patients get a notification once prescription is ready
- Basic view and listing of prescriptions for patients

Week 4 — Full Prescription Flow

- Approval step: doctor confirms before patient can see it
- Patients can download their prescription as a PDF
- System tracks expiry date and renewal date
- Keeps a history of all prescription changes

How It Works:

1. Doctor fills form
2. Prescription is saved
3. Doctor approves
4. Patient notified
5. Patient downloads PDF

Milestone 3 — Week 5

Medication Tracker & Reminder System

Add Medicines

- Patient enters medicine name and dosage
- Chooses how often to take it (daily, every 2 days, custom)
- Picks the time of day for each dose
- System saves the schedule automatically

Reminders

- App sends email or push notification at dose time
- Patient can tap 'Taken' or 'Snooze' from dashboard
- Missed doses are flagged for tracking
- Reminders repeat until patient responds

Adherence Tracking

- Shows % of doses taken on time
- Weekly and monthly trend view
- Helps patients stay on track
- Doctor can monitor patients

Auto Alert to Doctor

If a patient's adherence drops below the set limit (e.g. below 70%), the system automatically sends an alert to the doctor. The doctor can then follow up with the patient and update the treatment plan if needed.

Milestone 4 — Weeks 6 & 7

Pharmacy Inventory Management

Week 6 — Drug Inventory

- Pharmacist adds medicines to stock
- Each entry: name, batch no., expiry date, quantity
- Edit or delete medicine entries easily
- Search drugs by name quickly

Stock Alerts

- Auto alert when stock goes below set limit
- Expired medicines get flagged in red
- Pharmacist gets notification to restock
- Marks expired items for disposal

Week 7 — Drug API

- Fetches drug composition details
- Shows possible drug interactions
- Helps pharmacist do safety checks

Sample Inventory Table:

Drug Name	Batch No.	Expiry	Qty	Status
Paracetamol 500mg	BT-1021	Dec 2026	200	**OK**
Amoxicillin 250mg	BT-3045	Mar 2026	15	**Low Stock**
Metformin 500mg	BT-2210	Jan 2026	80	**Expired**

Milestone 5 — Week 8

Analytics & Reporting Dashboard

Dashboard visualization:

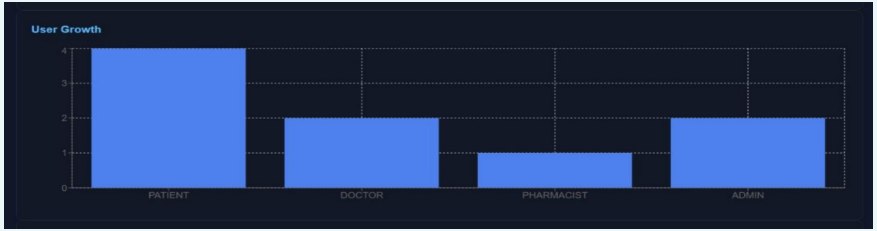
- Graphs for patients showing adherence trends over time.
- Doctor analytics for patient prescriptions issued and adherence levels

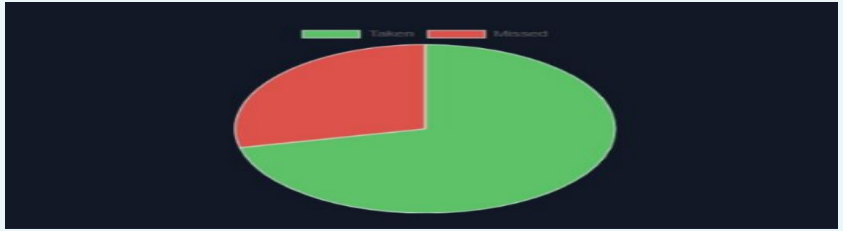
Pharmacy insights:

- Reports showing top-selling drugs and upcoming expirations.
- Sales vs. stock visualization for inventory optimization.

Overall system analytics:

- Admin can monitor total users, prescriptions, and alert logs.
- Generate downloadable reports for compliance and audits.





System Architecture

How all parts of the system connect

FRONTEND

React App

- Login & Signup
- Prescription View
- Medication Tracker
- Inventory Screen
- Analytics Charts

REST API

BACKEND — Spring Boot

- JWT Authentication
- Prescription Module
- Medication Tracker
- Pharmacy Inventory
- Analytics & Reports
- Notification Service

SQL

DATABASE — MySQL

- users
- prescriptions
- medicines
- reminders
- inventory
- reports

Project Summary

- Built a full-stack web app using React, Spring Boot, and MySQL
- Supports 4 user roles: Patient, Doctor, Pharmacist, and Admin
- Patients can track medicines and get reminders so they never miss a dose
- Doctors can issue and manage digital prescriptions easily
- Pharmacists can manage drug stock and get low-stock alerts
- Analytics dashboard gives useful reports for all user roles

Future Enhancements: Mobile App | AI Drug Interaction Checker | Insurance Integration | Telemedicine Feature

Thank You!